RICHA DUBEY

Ph.D. Scholar in Electrical Engineering
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EDUCATION

September 2020 – April 2026

Ph. D. (Thesis submitted) in Electrical Engineering
Indian Institute of Technology (IIT) Jodhpur

Main contributions of the thesis titled '*Data-Driven Anomaly Detection and Invariant Set-Based Resilient Control for Robots*' are

- Developed scalar and matrix Chernoff bound-based anomaly detection methods for robots to identify perturbations, feedback delays, and sensor corruption without heuristic threshold tuning.
- Designed decentralized and centralized real-time detection schemes using sliding-window deviation statistics for scalable robotic and multi-robot systems.
- Integrated Chernoff bound-based detection with Model Predictive Control (MPC) through asynchronous motion smoothing to reduce motion jitters in robots under perturbations.
- Formulated a finite-time MPC framework using λ-control invariant sets and derived explicit upper bounds on convergence time for predictable task execution.
- Combined corrupt sensor detection and identification with λ-robust control invariant sets to enable resilient control for robots under time-varying sensor corruption.
- Validated the proposed frameworks through simulations and hardware experiments on multiple robotic platforms.

August 2014 - July 2018

B. Tech. (Hons.) in Electrical Engineering
National Institute of Technology (NIT) Raipur
8.29/10 CGPA

2014
Kendriya Vidyalaya No.1 Raipur

Higher Secondary Certificate **(HSC)**
94.8% (CBSE Board)

2012
Kendriya Vidyalaya No.1 Raipur

Secondary School Certificate **(SSC)**
95% (CBSE Board)

PUBLICATIONS

Conference Publications

- **Dubey, R.**, Gupta, S., Chaudhary, S., Tripathy, N.S. and Shah, S.V., **2024,** August. *Finite-Time Convergence of Multi-robot Segregation using MPC with Aperiodic Motion Smoothing*. In IEEE 20th International Conference on Automation Science and Engineering (CASE) at Bari, Italy.

Journals Publications

- **Dubey, R.,** Tripathy, N.S. and Shah, S.V., *Data-Driven Anomaly Detection in Robots Using Matrix Chernoff Bounds*. IEEE Robotics and Automation Letters, Vol. 11, No. 3, March **2026**. (To be presented at ICRA 2026 in Vienna, Austria)
- **Dubey, R.,** Gupta, S., Chaudhary, S., Tripathy, N.S. and Shah, S.V., *Multi-Robot Segregation Using Finite-Time MPC With Chernoff Bound-Based Asynchronous Motion Smoothing*. IEEE Robotics and Automation Letters, Vol. 10, No. 11, November **2025**. (To be presented at ICRA 2026 in Vienna, Austria)
- **Dubey, R.**, Sharma, S. and Chaudhury, S., **2024**. *A Bibliometric Survey on Impact of Blockchain in Robotics: Trends and Applications*. Computers and Electrical Engineering.
- Chaudhary, S., **Dubey, R.**, Tripathy, N.S. and Shah, S.V., **2025**. *Data-Driven Event-Triggered Predictive Post-Impact Control of Space Robot with Uncertainties*. Journal of Guidance, Control, and Dynamics.

Manuscripts under preparation

- **Dubey, R.,** Tripathy, N.S. and Shah, S.V., *On the Detection and Mitigation of Sensor Corruption Using Matrix Chernoff Bounds and Invariant Sets*. (To be submitted to IEEE Transactions on Automation Science and Engineering).
- **Dubey, R.,** Tripathy, N.S. and Shah, S.V., *Set Invariance-Based Finite-Time Convergence of Robots Under Motion Smoothing and Feedback Delays*. (To be submitted to IEEE Transactions on Automatic Control).

GRANTS

August 2024
IEEE RAS Travel Grant

Awarded for presenting research work at IEEE 20th International Conference on Automation Science and Engineering (**CASE 2024**) in **Bari, Italy**.

September 2024
IEEE RAS Travel Grant

Awarded for presenting my work at the 40th Anniversary of IEEE International Conference on Robotics and Automation (**ICRA@40**) in **Rotterdam, Netherlands**.

April 2026
IEEE RAS Travel Grant

Awarded for presenting my work at the 2026 IEEE International Conference on Robotics and Automation (**ICRA**) in **Vienna, Austria.**

EXPERIENCE

2018 – 2019
Wipro Technologies

Project Engineer
Worked on forward compatibility of 3G with 4G technologies, gaining hands-on experience with UMTS networks, C, and UNIX at Wipro Technologies, Bengaluru, India.

May 2017 - June 2017
ISAC ISRO

Vocational Trainee
Studied HVAC systems under the C&M Group at ISRO Satellite Centre (ISAC), Government of India, Department of Space, Bengaluru, India.

2016
INDQ Innovations

R&D Intern
Worked in a student team developing assistive devices for visually impaired users at the National Institute of Design, Ahmedabad, India.

ROLES AND RESPONSIBILITIES

September 2020 - July 2025
Department of Electrical Engineering
IIT Jodhpur

Teaching Assistant
Assisted in the following courses: Control Systems, Advanced Control Systems, Control Systems Laboratory, Digital Systems Laboratory

- Conducted laboratory sessions for approximately 25 undergraduate students; helped them learn coding, hardware-software interfacing, and the connection between theory and real-time experiments.
- Assisted in lab examinations, viva voce, evaluations and grading.
- Conduced tutorial sessions for Master’s students, addressed their doubts and queries, brainstormed ideas with them for projects.

2024 - Present

Reviewer
Conferences: *IEEE IROS, IEEE CASE, Advances in Robotics (AIR)*
Journals: *IEEE Robotics and Automation Letters (RA-L), Springer Nature Journal of Control, Automation and Electrical Systems*

July 2025
7th International Conference on Advances in Robotics (AIR)

Master of Ceremonies
Led a team of four hosts for a three-day conference; wrote and supervised scripts, and coordinated with speakers to ensure the timely and effective conduct of the conference.

March 2024
Workshop on Field Robotics
IIT Jodhpur

Event Coordinator
Served as Master of Ceremonies and Event Coordinator for the Workshop on Field Robotics, organized by IIT Jodhpur and The Robotics Society of India.

July 2016 - July 2018
The Placement Office
NIT Raipur

Placement Convener
Coordinated and assisted placement drives at NIT Raipur for various companies and organizations

SKILLS

Technical Skills

- MATLAB, Simulink, C, C++, Python, LaTeX, ROS
- MS Word, Excel, PowerPoint
- Data Science and Machine Learning certifications from Udemy

Transferable Skills

Collaborative Research and Teamwork
Worked in multidisciplinary research teams, yielding publications that explore novel directions in advancements of robotics research.

Public Speaking and Oral Communication
Served as Master of Ceremonies for academic conferences and institutional events; presented research work at national and international conferences and doctoral symposia.

Technical and Creative Writing
Wrote and edited survey papers, conference papers, journal articles, and anchoring scripts for academic events; also engaged in poetry writing as a creative pursuit.

Event Organizing and Team Management
Worked in organizing teams for national and international conferences and workshops, including coordination of volunteers.

Languages

English (Full Professional), Hindi (Native), French (Beginner)

REFEREES

Dr. Niladri S. Tripathy
(Ph.D. Supervisor)

Assistant Professor, Department of Electrical Engineering, IIT Jodhpur
niladri@iitj.ac.in, Ph.: 0291 280 1375

Prof. Suril V. Shah
(Ph.D. Co-Supervisor)

Professor, Department of Mechanical Engineering, IIT Jodhpur
surilshah@iitj.ac.in, Ph.: 0291 280 1510